JanGyan — Learn Without Limits

Category 4: Health & Emergency Awareness

Module 5 - Communicable & Lifestyle Diseases

Module Overview

India faces a unique "double burden" of disease — infectious illnesses like TB, dengue, and malaria remain widespread, even as lifestyle conditions like diabetes, hypertension, and obesity rise sharply with changing habits. This handbook is Module 5 of JanGyan's Health & Emergency Awareness category, and it equips every citizen to understand, prevent, and manage the diseases most likely to affect Indian families today.

By the end of this handbook, you will understand how infections spread and how to prevent them, recognise the warning signs of TB, dengue, malaria, and COVID-19, understand what diabetes, hypertension, and obesity really mean for your body, and know exactly how India's vaccination and screening programmes can protect you and your family. This module is designed as a complete, self-sufficient guide.

Learning Objectives

After completing this module, you will be able to:

- Distinguish communicable from non-communicable (lifestyle) diseases.
- Explain how infections spread from person to person.
- Recognise the warning signs of TB, dengue, malaria, and COVID-19.
- Understand the basics of viral and bacterial infections, and why antibiotics don't treat viruses.
- Recognise the risk factors and warning signs of diabetes, hypertension, and obesity.
- Explain the role of vaccination in preventing disease.
- Navigate India's key national disease-control programmes.
- Separate common disease-related myths from medically accurate facts.

Why This Topic Matters

Important

India carries a "double burden" of disease: infectious illnesses have not disappeared even as lifestyle diseases surge. The ICMR-INDIAB study estimates that around 101 million Indians live with diabetes and roughly 315 million with hypertension — while TB alone still affects millions each year, making India home to the world's largest TB burden.

Many of these conditions are highly preventable or manageable with basic awareness — knowing that TB spreads through the air and is curable with free government treatment, that dengue has no specific cure but is preventable by controlling mosquito breeding, or that hypertension often has no symptoms at all until it causes serious harm.

This handbook exists to close exactly that awareness gap — the space between "not knowing" and "knowing what to do."

Introduction

Diseases are broadly grouped into two families: communicable diseases, which spread from person to person or via a vector like a mosquito, and non-communicable diseases, which develop gradually due to lifestyle, genetics, or environment and do not spread between people.

Definition

A **Communicable Disease** is an illness caused by an infectious agent (virus, bacteria, or parasite) that can be transmitted from one person, animal, or vector to another.

Definition

A **Non-Communicable Disease (NCD)**, sometimes called a lifestyle disease, is a long-term condition that is not passed between people, typically linked to diet, physical activity, and other lifestyle factors.

This module covers both families side by side, because in India today, they are often managed by the same public health system — and frequently affect the same households.

Fundamental Concepts

1. Communicable vs Non-Communicable Diseases

Aspect	Communicable Diseases	Non-Communicable Diseases
Cause	Virus, bacteria, parasite	Lifestyle, genetics, environment
Spreads between people?	Yes	No
Examples	TB, dengue, malaria, COVID-19	Diabetes, hypertension, obesity
Onset	Often sudden	Usually gradual
Prevention Focus	Hygiene, vaccination, vector control	Diet, activity, regular screening

2. How Infections Spread

Mode of Spread	Example
Airborne / Droplet	TB, COVID-19, common cold — spread through coughs, sneezes, or shared air
Vector-borne	Dengue, malaria — spread through mosquito bites
Contact	Skin infections — spread through touch or contaminated surfaces
Waterborne / Foodborne	Diarrhoeal disease — spread through contaminated water or food

Remember

Breaking any one link in this chain — covering a cough, eliminating mosquito breeding spots, washing hands — is often enough to stop an infection from spreading further.

Complete Detailed Explanation

Each disease below follows the same structure: what it is, its warning signs, and how it is prevented or managed.

3. Viral Infections

Viral infections (like the common cold, flu, or viral hepatitis) are caused by viruses — tiny agents that need a living cell to multiply. Most are self-limiting, meaning the body's immune system clears them on its own over time.

- **Typical signs:** Fever, fatigue, body ache, cough or cold symptoms.
- **Management:** Rest, fluids, and symptom relief; most resolve within days to two weeks.

Common Mistake

Antibiotics do not work against viruses. Taking them for a viral infection doesn't help you recover faster and contributes to antibiotic resistance.

4. Bacterial Infections

Bacterial infections are caused by bacteria and can often be treated effectively with the correct antibiotic, prescribed by a doctor.

- **Typical signs:** Localised pain, swelling, fever, or pus, depending on the site of infection.
- **Management:** A full prescribed course of antibiotics, even after symptoms improve.

Warning

Stopping antibiotics early, or using leftover antibiotics without a prescription, helps bacteria develop resistance — a growing public health concern in India and worldwide.

5. Tuberculosis (TB)

TB is a bacterial infection, usually affecting the lungs, that spreads through the air when an infected person coughs or sneezes. India carries the world's largest TB burden, but TB is both preventable and completely curable with free treatment.

Aspect	Detail
Warning Signs	Cough lasting more than 2 weeks, fever, unexplained weight loss, night sweats
Treatment	Free, government-supplied multi-drug course, typically lasting 6 months or more
Government Programme	National TB Elimination Programme (NTEP), targeting TB elimination in India by 2025

Indian Context

Under the Pradhan Mantri TB Mukt Bharat Abhiyaan, community volunteers called "Ni-Kshay Mitras" support TB patients with nutrition and care, while the Ni-Kshay Poshan Yojana provides direct financial support to registered patients during treatment.

6. Dengue

Dengue is a viral infection spread by the Aedes mosquito, which typically bites during the day and breeds in clean, stagnant water around homes.

- **Warning Signs:** High fever, severe headache, pain behind the eyes, joint and muscle pain, rash.
- **Management:** There is no specific antiviral cure — treatment focuses on rest, fluids, and monitoring; severe cases need hospital care.
- **Prevention:** Eliminate stagnant water around homes, use mosquito nets and repellents.

7. Malaria

Malaria is a parasitic infection spread by the Anopheles mosquito, which typically bites between dusk and dawn.

- **Warning Signs:** Cyclical fever with chills and sweating, headache, nausea.
- **Management:** Confirmed by a blood test and treated with specific anti-malarial medication.

> **Indian Context**
> India's National Framework for Malaria Elimination (2016–2030) has driven a 76% drop in confirmed malaria cases between 2016 and 2024, and India exited the WHO's "High Burden to High Impact" group in 2024 — a genuine public health success story.

8. COVID-19 Awareness

COVID-19, caused by the SARS-CoV-2 virus, is no longer classified by the WHO as a global health emergency — it is now managed as an ongoing respiratory illness, similar to seasonal influenza, that continues to circulate and evolve.

- **Typical Signs:** Fever, cough, sore throat, fatigue, loss of taste or smell in some cases.
- **Precautions:** Test when symptomatic, stay updated on vaccination if you are high-risk (older adults or those with underlying conditions), maintain hand hygiene, and ventilate indoor spaces.

> **Tip**
> Vaccine recommendations continue to evolve as the virus does. High-risk individuals should check current government guidance on booster eligibility rather than relying on outdated information.

9. Diabetes

Diabetes is a condition where blood sugar levels remain too high, either because the body doesn't produce enough insulin (Type 1) or doesn't use it effectively (Type 2, the far more common form in adults).

- **Warning Signs:** Excessive thirst, frequent urination, unexplained fatigue, slow-healing wounds.
- **Management:** Balanced diet, regular physical activity, medication as prescribed, and regular blood sugar monitoring.

> **Statistics**
> The ICMR-INDIAB study estimates that 11.4% of Indian adults have diabetes and a further 15.3% have prediabetes — meaning early lifestyle changes could prevent many future cases.

10. Hypertension

Hypertension (high blood pressure) is often called the "silent killer" because it usually causes no symptoms at all, even while quietly damaging the heart, kidneys, and blood vessels.

- **Warning Signs:** Often none — regular checking is the only reliable way to detect it.

- **Management:** Reduced salt intake, regular exercise, stress management, and medication as prescribed.

> **Statistics**
> The ICMR-INDIAB study found hypertension in 35.5% of Indian adults — an estimated 315 million people — making regular blood pressure checks one of the simplest, highest-value health habits available.

11. Obesity

Obesity is excess body fat that increases the risk of diabetes, hypertension, heart disease, and joint problems. It is commonly assessed using Body Mass Index (BMI) and waist circumference.

- **Warning Signs:** BMI above 25 (overweight) or 30 (obese) under WHO Asia-Pacific guidelines used in India; increased waist circumference.
- **Management:** Balanced nutrition, regular physical activity, and gradual, sustainable weight management rather than crash diets.

12. Prevention & Vaccination

India's Universal Immunisation Programme (UIP), running since 1985, is among the largest vaccination programmes in the world, protecting against 12 vaccine-preventable diseases including TB, measles, and polio.

Programme	Key Fact
Universal Immunisation Programme (UIP)	Free vaccines for children and pregnant women against 12 diseases
Mission Indradhanush	Launched 2014 to push full immunisation coverage above 90% in low-coverage areas
U-WIN	Digital platform tracking individual immunisation records in real time

> **Statistics**
> India's full child immunisation coverage has risen from 62% in 2015 to over 98% in early 2026 — one of the most significant public health improvements in the country's recent history.

Important Terminologies

Term	Simple Meaning
Pathogen	A microorganism (virus, bacteria, parasite) that causes disease.
Vector	A living organism, like a mosquito, that transmits disease between hosts.
Antibiotic Resistance	When bacteria evolve to survive medicines that once killed them.
Immunisation	Building protection against a disease, usually through vaccination.
BMI	Body Mass Index — a simple ratio of weight to height used to screen for healthy body weight.
Prediabetes	Blood sugar levels higher than normal, but not yet in the diabetic range.
Silent Killer	A term for conditions like hypertension that cause serious harm with no early symptoms.
DOTS	Directly Observed Treatment, Short-course — the supervised TB treatment approach used in India.

Step-by-Step Understanding

Responding to a Suspected Infectious Illness

1. **Observe:** Note the specific symptoms and how long they've lasted.
2. **Isolate sensibly:** Limit close contact with others if a contagious illness is suspected.
3. **Seek testing:** Visit a government health facility or doctor for accurate diagnosis.
4. **Follow treatment fully:** Complete the entire prescribed course, especially for TB or bacterial infections.
5. **Prevent spread:** Practise hand hygiene, cover coughs, and manage mosquito breeding sites at home.

Building a Lifestyle Disease Prevention Routine

1. Get blood pressure and blood sugar checked at least once a year after age 30.
2. Build regular physical activity into your daily routine.
3. Choose home-cooked, balanced meals over processed food where possible.
4. Track your BMI and waist circumference periodically.
5. Visit your nearest Ayushman Arogya Mandir (Health and Wellness Centre) for free NCD screening.

Real-Life Indian Examples

Example 1

A textile worker in Surat diagnosed with TB completed his full 6-month treatment with support from a Ni-Kshay Mitra volunteer, who provided monthly nutrition kits — he made a full recovery and now encourages colleagues to get tested early.

Example 2

A housing society in Pune organised a weekly "dry day" to empty stagnant water from coolers and plant pots, sharply reducing dengue cases in the neighbourhood the following monsoon season.

Example 3

A 42-year-old teacher in Kochi discovered she had hypertension only during a routine check at her workplace health camp — she had no symptoms at all, but early treatment prevented more serious complications.

Example 4

A father in Lucknow who once delayed his son's vaccines due to a busy work schedule caught up fully after a health worker explained Mission Indradhanush's catch-up vaccination options at the local Anganwadi centre.

Practical Scenarios

Scenario	What You Should Do
You've had a cough for over 2 weeks	Get tested for TB at a government health facility — testing and treatment are both free.

A family member has high fever with pain behind the eyes	Consult a doctor promptly to rule out dengue; avoid self-medicating with certain painkillers without medical advice.
Your blood pressure reading is high but you feel completely fine	Take it seriously anyway — hypertension often has no symptoms; follow up with a doctor.
You finish half a course of antibiotics because you feel better	Complete the full course as prescribed to avoid contributing to antibiotic resistance.
Your child has missed a scheduled vaccine dose	Visit your nearest health centre for catch-up vaccination — it's never too late to complete the schedule.

Government Rules & Guidelines

Programme	Focus
National TB Elimination Programme (NTEP)	Free TB diagnosis and treatment nationwide, targeting elimination by 2025
National Centre for Vector Borne Disease Control (NCVBDC)	Control of malaria, dengue, and four other vector-borne diseases
National Programme for Prevention & Control of Cancer, Diabetes, CVD & Stroke (NPCDCS)	Screening and management of major lifestyle diseases
Universal Immunisation Programme (UIP) / Mission Indradhanush	Free vaccination for children and pregnant women nationwide
Ayushman Arogya Mandirs (Health & Wellness Centres)	Free screening for common NCDs, close to where people live

Laws & Policies

India's disease-control efforts are guided by the **National Health Policy, 2017**, which emphasises both communicable disease control and the growing burden of non-communicable diseases as national priorities. TB notification is mandatory under Indian law — every diagnosed case must be reported to the National Tuberculosis Elimination Programme's Ni-kshay portal, helping the government track and support every patient.

Good to Know

Because TB treatment and reporting are free and mandatory under government policy, patients face no legal or financial barrier to completing treatment — cost should never be a reason to stop.

Important Statistics

Statistic	Insight
An estimated 101 million Indian adults have diabetes (ICMR-INDIAB)	India carries one of the largest diabetes populations in the world.
An estimated 315 million Indian adults have hypertension (ICMR-INDIAB)	Roughly 1 in 3 adults — most without knowing it.

TB incidence in India fell 21% between 2015 and 2024 (WHO Global TB Report 2025)	Shows real progress, though India still carries the world's largest TB burden.
Malaria cases fell 76% between 2016 and 2024 (NCVBDC)	One of India's clearest public health success stories.
Full child immunisation coverage rose from 62% (2015) to over 98% (2026)	Reflects the impact of Mission Indradhanush and the wider UIP.

Suggested Visual: A side-by-side bar chart comparing India's communicable vs lifestyle disease burden over the last decade.

Best Practices

- Complete every prescribed course of antibiotics or TB medication fully.
- Get blood pressure and blood sugar screened at least annually after age 30.
- Eliminate stagnant water around your home weekly to prevent mosquito breeding.
- Keep your family's vaccination schedule up to date, including catch-up doses.
- Choose balanced, home-cooked meals and stay physically active most days.
- Visit your nearest Ayushman Arogya Mandir for free NCD screening.

Common Mistakes

- Taking antibiotics for viral infections like the common cold or flu.
- Stopping TB or antibiotic treatment early once symptoms improve.
- Assuming no symptoms means no hypertension or diabetes risk.
- Ignoring a cough lasting more than two weeks.
- Delaying children's vaccines due to minor illness or scheduling difficulty.
- Self-medicating for suspected dengue instead of consulting a doctor.

Myths vs Facts

Myth	Fact
"Antibiotics cure the common cold faster."	Colds are viral — antibiotics have no effect on them.
"TB only affects poor or malnourished people."	TB can affect anyone exposed to the bacteria, regardless of economic status.
"If I feel fine, my blood pressure must be normal."	Hypertension usually has no symptoms — only a check can confirm it.
"Diabetes only affects overweight people."	Diabetes can affect people of any body type, including those of normal weight.
"Once a vaccine schedule is missed, it's too late to catch up."	Catch-up vaccination is available and effective at nearly any age.

Do's and Don'ts

Do	Don't
Complete the full course of any prescribed antibiotic or TB medication	Don't stop medication early just because you feel better
Get regular blood pressure and blood sugar checks	Don't assume you're healthy just because you feel fine
Empty stagnant water around your home weekly	Don't ignore small pools of standing water indoors or outdoors
Keep vaccination records and schedules up to date	Don't delay a child's vaccine over a minor cold or fever alone
Consult a doctor for fever lasting more than a couple of days	Don't self-medicate for suspected dengue or malaria

Safety Tips

- Cover your mouth and nose when coughing or sneezing to protect others.
- Wash hands regularly, especially before eating and after using the toilet.
- Use mosquito nets or repellents during dengue and malaria season.
- Keep prescribed medication for chronic conditions like diabetes or hypertension on schedule.
- Don't hesitate to seek a second medical opinion for a serious or unclear diagnosis.

Practical Checklist

Personal Health Screening Checklist

☐ Blood pressure checked in the last year

☐ Blood sugar checked in the last year (especially after age 30)

☐ Family vaccination schedule up to date

☐ No stagnant water sources around the home

☐ Aware of nearest Ayushman Arogya Mandir / Health & Wellness Centre

☐ Any cough lasting over 2 weeks has been checked by a doctor

Quick Revision Notes

Revision

- Communicable diseases spread between people; non-communicable diseases don't.
- TB: cough >2 weeks, free government treatment, must be completed fully.
- Dengue and malaria: mosquito-borne, no specific antiviral cure, prevention is key.
- Hypertension and diabetes often have no symptoms — screening is essential.
- Antibiotics treat bacteria, not viruses.
- Vaccination remains one of the most effective, affordable prevention tools available.

Key Takeaways

Key Takeaway

1. India faces both infectious and lifestyle diseases at the same time — awareness of both matters.
2. Many serious conditions, like TB and hypertension, are highly treatable if caught early.
3. Simple habits — hand hygiene, mosquito control, regular screening — prevent enormous harm.
4. Government programmes make TB treatment, vaccination, and NCD screening free and accessible.
5. "Feeling fine" is not the same as "being healthy" — regular checks matter.

Glossary

Pathogen — A microorganism that causes disease.

Vector — A living organism, such as a mosquito, that transmits disease.

NCD — Non-Communicable Disease, such as diabetes or hypertension.

BMI — Body Mass Index, a screening tool for healthy body weight.

DOTS — Directly Observed Treatment, Short-course, used for TB treatment supervision.

Prediabetes — Blood sugar higher than normal but below the diabetic threshold.

Immunisation — Building protection against disease, usually through vaccines.

Frequently Asked Questions

1. What is the difference between communicable and non-communicable diseases?

Communicable diseases spread between people or via a vector; non-communicable diseases develop from lifestyle or genetic factors and don't spread.

2. Why don't antibiotics work on viral infections?

Antibiotics target bacteria; viruses have a completely different structure that antibiotics cannot act on.

3. What is the main warning sign of TB?

A cough lasting more than 2 weeks, often with fever, weight loss, and night sweats.

4. Is TB treatment free in India?

Yes — diagnosis and treatment are provided free under the National TB Elimination Programme.

5. How is dengue spread?

Through the bite of an infected Aedes mosquito, which breeds in clean, stagnant water.

6. Is there a specific cure for dengue?

No — treatment focuses on rest, fluids, and monitoring, with hospital care for severe cases.

7. How is malaria spread?

Through the bite of an infected Anopheles mosquito, typically between dusk and dawn.

8. Is COVID-19 still a global emergency?

No — the WHO now manages it as an ongoing respiratory illness rather than a global health emergency.

9. What are the warning signs of diabetes?

Excessive thirst, frequent urination, unexplained fatigue, and slow-healing wounds.

10. Why is hypertension called a "silent killer"?

Because it usually causes no symptoms, even while damaging the heart and blood vessels over time.

11. How is obesity measured?

Commonly through Body Mass Index (BMI) and waist circumference.

12. What is Mission Indradhanush?

A government initiative launched in 2014 to raise full child immunisation coverage above 90%, focused on under-reached areas.

13. How many diseases does India's Universal Immunisation Programme cover?

12 vaccine-preventable diseases, including TB, measles, and polio.

14. What should I do if I have a cough lasting more than two weeks?

Get tested for TB at a government health facility — testing and treatment are free.

15. Can diabetes affect people who are not overweight?

Yes — body type alone does not determine diabetes risk.

16. What percentage of Indian adults have hypertension?

An estimated 35.5%, according to the ICMR-INDIAB study.

17. Why is completing a full antibiotic course important?

Stopping early can allow surviving bacteria to develop resistance to the medicine.

18. What is a Ni-Kshay Mitra?

A community volunteer who supports TB patients with nutrition and care under the PM TB Mukt Bharat Abhiyaan.

19. Where can I get free NCD screening in India?

At your nearest Ayushman Arogya Mandir (Health and Wellness Centre).

20. Is it too late to vaccinate a child who missed scheduled doses?

No — catch-up vaccination is available and remains effective.

21. What does "prediabetes" mean?

Blood sugar levels higher than normal but not yet high enough to be classified as diabetes.

22. How much has malaria declined in India recently?

Confirmed cases fell by 76% between 2016 and 2024.

23. What is DOTS in TB treatment?

Directly Observed Treatment, Short-course — a supervised approach ensuring patients complete their medication correctly.

24. Should I worry about a high BMI even without other symptoms?

Yes — obesity raises the risk of diabetes, hypertension, and heart disease even without immediate symptoms.

25. What is the single best defence against most of the diseases in this handbook?

A combination of hygiene, vector control, timely vaccination, and regular health screening.

Suggested Infographics, Flowcharts & Tables

- Infographic: Communicable vs Non-Communicable diseases, shown side by side.
- Flowchart: How an infection spreads from source to new host.
- Poster: TB warning signs and the free treatment pathway.
- Chart: India's declining malaria and TB trends over the last decade.
- Card: BMI categories and what they mean for health risk.

Official References

- Ministry of Health & Family Welfare, Government of India — National TB Elimination Programme (NTEP) and National Strategic Plan 2017–2025.
- National Centre for Vector Borne Disease Control (NCVBDC) — malaria and dengue surveillance data.
- Indian Council of Medical Research (ICMR) — the ICMR-INDIAB national study on diabetes, hypertension, and obesity.
- World Health Organization (WHO) — Global Tuberculosis Report 2025 and COVID-19 guidance updates.
- Ministry of Health & Family Welfare — Universal Immunisation Programme and Mission Indradhanush.
- National Health Policy, 2017, Government of India.

Official Government Website Links

Resource	Website
Central TB Division (NTEP)	tbcindia.mohfw.gov.in
National Centre for Vector Borne Disease Control	ncvbdc.mohfw.gov.in
Ministry of Health & Family Welfare	mohfw.gov.in
National Health Mission	nhm.gov.in
Indian Council of Medical Research	icmr.gov.in

Conclusion

Whether it's an infection spread through a mosquito bite or a lifestyle condition that builds up silently over years, the pattern is the same: awareness and early action change outcomes dramatically. TB is curable, dengue and malaria are preventable, and hypertension and diabetes are manageable — but only if they are recognised and acted on in time.

As you move ahead in the JanGyan platform, use the Practice and Quiz sections to reinforce what you've learned here, and don't hesitate to use AI Assistance whenever a doubt arises. Understanding these diseases is not just personal protection — it is a contribution to a healthier community around you.

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Module 5 of the Health & Emergency Awareness Category